ADB

Extended Annual Review Report

PUBLIC

Project Number: 50200-001
Loan Number: 3448
November 2022

Triconboston Consulting Corporation (Private) Limited
Triconboston Wind Power Project
(Pakistan)

This is a redacted version of the document, which excludes information that is subject to exceptions to disclosure set forth in ADB's Access to Information Policy.

Asian Development Bank

CURRENCY EQUIVALENTS

Currency unit – Pakistan rupee/s (PRe/PRs)

		At Appraisal	At Operations Evaluation
		(15 September 2016)	(31 March 2022)
PRe1.00	–	$0.0096	$0.0055
$1.00	–	PRs104.58	PRs182.58

ABBREVIATIONS

ADB	–	Asian Development Bank
BAFL	–	Bank Alfalah Limited
COVID-19	–	coronavirus disease
DEG	–	Deutsche Investitions- und Entwicklungsgesellschaft
DFI	–	development financial institution
DMF	–	design and monitoring framework
DSO	–	days sales outstanding
E&S	–	environmental and social
EPA	–	energy purchase agreement
EPC	–	engineering, procurement, and construction
FiT	–	feed-in tariff
IEE	–	initial environmental examination
IFC	–	International Finance Corporation
IPP	–	independent power producer
NEPRA	–	National Electric Power and Regulatory Authority
NPMV	–	non-project missed volumes
NTDC	–	National Transmission and Despatch Company Limited
O&M	–	operation and maintenance
RRP	–	report and recommendation of the President
STML	–	Sapphire Textile Mills Limited
TBCC	–	Triconboston Consulting Corporation (Private) Limited
TWPP	–	Triconboston Wind Power Project
XARR	–	extended annual review report

WEIGHTS AND MEASURES

GWh	–	gigawatt-hour
km	–	kilometer
kWh	–	kilowatt-hour
MW	–	megawatt

NOTES

(i) The fiscal year (FY) of Triconboston Consulting Corporation (Private) Limited ends on 30 June of the respective year; e.g., FY2018 covers 1 July 2017 to 30 June 2018.

(ii) In this report, "$" refers to United States dollars.

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BASIC DATA
Triconboston Wind Power Project
(Loan number 3448 – Pakistan)

Key Project Data	As per RRP ($ million)	Actual ($ million)
Total project cost	**360.0**	**316.8**
Equity (including shareholder's loan)	**90.0**	**79.2**
Debt:	**270.0**	**237.6**
Asian Development Bank	75.0	66.0
Deutsche Investitions- und Entwicklungsgesellschaft	45.0	39.6
International Finance Corporation	75.0	66.0
Islamic Development Bank	75.0	66.0

RRP = report and recommendation of the President.

Key Dates	Expected	Actual
Concept Clearance Approval	6 June 2016	8 June 2016
Fact Finding Missions	not applicable	June–October 2016
Board Approval		27 October 2016
Loan Agreement	30 November 2016	10 April 2017
First Disbursement		1 August 2017
Commercial Operations Date	30 November 2018	14 September 2018

Project Administration and Monitoring	Number of Missions	Number of Person-Days
Due Diligence and Appraisal	4	28
Project Administration	1	4
XARR Mission[a]	0	0

XARR = extended annual review report.
[a] No XARR mission was scheduled because of travel restrictions arising from the coronavirus disease (COVID-19) pandemic.

EXECUTIVE SUMMARY

On 27 October 2016 the Board of Directors of the Asian Development Bank (ADB) approved an 11-year term loan of $75.0 million to Triconboston Consulting Corporation (Private) Limited (TBCC) for the financing of three 50-megawatt (MW) wind power plants in Pakistan.

The Triconboston Wind Power Project (TWPP) was developed following an ongoing initiative of the Government of Pakistan to promote private sector participation in the country's energy infrastructure. In addition, Pakistan urgently needed to develop indigenous energy resources to reduce its dependence on imported fuel, which accounted for a third of its generation capacity.

With an installed capacity of 150 MW, the TWPP is the largest single wind project in the country. It is located in the Gharo–Jhimpir wind corridor in Sindh Province, about 100 kilometers northeast of Karachi.

The project has been evaluated for (i) its development results, (ii) ADB additionality, (iii) ADB investment profitability and (iv) ADB work quality. The development results of the project have been rated as *satisfactory* based on performance across the following four categories: (i) contributions to private sector development and ADB strategic development objectives; (ii) economic performance; (iii) environment, social, health, and safety performance; and (iv) business success.

The contributions to private sector development and ADB strategic development objectives are rated *satisfactory*. The TWPP instilled confidence in the wind power sector and catalyzed private sector participation. At the time of loan processing in 2016, the wind power sector had only 306 MW of installed generation capacity. As of 2021, the wind power sector had a total installed generation capacity of 1,248 MW. The TWPP has also contributed to various other aspects of the wind power sector's development. The project's objectives were consistent with (i) ADB's Midterm Review of Strategy 2020, (ii) ADB's country partnership strategy for Pakistan 2015–2019, and (iii) ADB's energy policy (2009), which prioritized investments to maximize access to energy for all. The project rationale was also consistent with the government's target to achieve 9,700 MW of renewable power by 2030.

The project's economic performance is rated *satisfactory*.

The project is rated *satisfactory* for environmental, social, health, and safety performance, based on a review and evaluation of available safeguard documents, interviews with the client, and a virtual site visit.

The project is rated *excellent* for business success.

ADB's additionality is rated *satisfactory*. ADB's involvement in the project (i) provided dollar-denominated long-term financing that was limited to infrastructure projects in Pakistan and (ii) reassured project sponsors to undertake a large project, which was also an early adopter of the new up-front tariff regime.

The project is rated *satisfactory* for ADB's investment profitability.

ADB's work quality is rated *satisfactory*. ADB's effectiveness in screening, appraisal, and structuring is rated *satisfactory*. The project is in line with ADB's country, energy sector, private sector development, and environmental protection strategies. Waiver requests have been addressed in a balanced manner.

Overall, the project is rated *successful*.

The project's main issues and lessons learned are that (i) pioneering projects have a high development impact; (ii) suitable sponsor selection is critical for success, especially for pioneering projects; and (iii) the catalytic impact of pioneering projects is more substantial than their immediate impacts.

ADB will continue to monitor the TWPP's energy generation, particularly tracking curtailments and their impact on the project's ability to achieve its targets for energy generation and greenhouse gas emissions avoidance. ADB will also continue regular communication with other development finance institutions as and when there are further developments regarding tariff renegotiations.

I. THE PROJECT

A. Project Background

1. On 27 October 2016 the Board of Directors of the Asian Development Bank (ADB) approved a $75.0 million 11-year term loan to Triconboston Consulting Corporation (Private) Limited (TBCC) for the financing of three 50-megawatt (MW) wind power plants in Pakistan (Tricon-A, -B, and -C).[1]

2. The Triconboston Wind Power Project (TWPP) was developed following the ongoing initiative of the Government of Pakistan to promote private sector participation in the country's energy infrastructure, to provide sustainable energy supply. Several energy policies had resulted in significant investment from the private sector. As of FY2015, private sector investments owned and operated 11,574 MW or 46.2% of the country's installed power capacity.[2] In addition, Pakistan urgently needed to develop indigenous energy resources to reduce its dependence on imported fuel, which accounted for a third of its generation capacity.

3. ADB had already financed three independent power producers (IPPs) of wind energy under the 2006 Renewable Energy Power Policy. The TWPP is ADB's fourth wind project in the country and to date remains the largest wind project in Pakistan.

B. Key Project Features

4. TBCC was established as a special purpose vehicle to develop, construct, commission, and operate three adjacent 50 MW wind power plants (Tricon-A, -B, and -C). The project is located in the Gharo–Jhimpir wind corridor in Sindh Province, about 100 kilometers (km) northeast of Karachi.

5. With a combined capacity of 150 MW, the TWPP is the largest single IPP of wind energy in Pakistan. The project was part of the second generation of wind IPPs, developed following predecessors such as Foundation Wind Energy I and II and Zorlu Enerji Pakistan Limited.

6. Each of the three plants is fitted with 29 General Electric 1.7 MW turbines. The project's total annual output (based on P90) was estimated at 520 gigawatt-hours (GWh), corresponding to a capacity factor of 39.6%.[3] Power is evacuated by the national grid operator, National Transmission and Despatch Company Limited (NTDC), over a 7 km transmission line.

7. The project was implemented under a fixed-rate, date-certain, turnkey engineering, procurement, and construction (EPC) arrangement and awarded through a competitive bidding process. HydroChina Corporation, a subsidiary of state-owned Power Construction Corporation of China, won and carried out the EPC contract.[4] General Electric supplied wind turbines.

8. Power offtake is secured through a 20-year take-or-pay energy purchase agreement (EPA) signed by each subproject with the NTDC, through its clearing agent, the Central Power Purchasing Agency. In turn, the government guarantees the NTDC's obligations under the EPA as part of its commitments under a 20-year Implementation Agreement.

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[1] ADB. 2016. *Report and Recommendation of the President to the Board of Directors on a Proposed Loan to Triconboston Consulting Corporation (Private) Limited for the Triconboston Wind Power Project. Manila.*

[2] National Electric Power Regulatory Authority. 2016. *State of the Industry Report 2015*. Islamabad.

[3] P90 denotes the level of annual wind speed where 90% of the possible estimates exceed the value assumed.

[4] At appraisal, HydroChina had EPC experience with more than 180 MW of capacity at the Jhimpir site.

9. The EPA allows for non-project missed volumes (NPMV), compensating the project for grid failures and curtailments. However, unlike earlier-vintage wind IPPs, the EPA did not include wind-speed risk guarantees, and resource risk was borne by the project.

10. Under the EPA, the project has an upfront feed-in tariff, providing a 17% return on equity and compensating for operation and maintenance and debt service (principal and interest). The tariff is indexed to allow for escalation in inflation, exchange rates, and the London interbank offered rate.

11. On 13 May 2016, the National Electric Power and Regulatory Authority (NEPRA) granted the TWPP approval for a levelized reference tariff of $0.1045 per kilowatt-hour (kWh) for the EPA's 20-year duration ($0.1242/kWh for the first 10 years and $0.566/kWh for the second 10-year period).[5]

12. The project cost was originally estimated at $360 million, to be financed with a 75:25 debt-equity ratio. Debt funding was arranged under a typical non-recourse project finance structure and comprised (i) a $75 million loan from ADB, (ii) a $45 million loan from Deutsche Investitions- und Entwicklungsgesellschaft, (iii) a $75 million loan from the International Finance Corporation, and (iv) a $75 million loan from the Islamic Development Bank. All loans have an 11-year tenor, including a 2-year grace period.

13. The TWPP's main sponsor is the Sapphire Group (78.5%),[6] a leading Pakistani conglomerate with business interests in textiles, food, and power (both thermal and renewables). It owns and operates a 53 MW wind farm in Jhimpir Sindh that began commercial operation in 2015. The Group also owns and operates a 234 MW combined-cycle thermal power project (commissioned in 2010). Other project sponsors include Bank Alfalah Limited (BAFL),[7] Pakistan's sixth largest bank (9.3%), and Baltoro Growth Fund (12.2%), one of the three funds created under the US Agency for International Development's Pakistan Private Investment Initiative.

C. Progress Highlights

14. The project's finance documents were signed on 10 April 2017, and financial close was achieved on 10 May 2017, following approval and registration of the finance documents with the State Bank of Pakistan, the registration of the EPC documents with the bank, and signing of the EPA as well as the Implementation Agreement, among other steps.

15. Preliminary enabling site works commenced in December 2016. The first disbursement occurred on 1 August 2017. Receipt of the initial notice to proceed was provided on 4 August

[5] Offtake payments are in Pakistan rupees (PRs). The EPA allows full indexation of tariffs to account for any fluctuations in dollar-denominated costs. These adjustments are based on pre-determined formula and reference parameters on a quarterly basis, as set out by the EPA with the CPPA.

[6] Founded in 1940, the Sapphire Group is one of Pakistan's largest conglomerates, a network of family-owned and -managed companies with investments in textiles, power, dairy, and real estate development. It had a turnover and asset base of almost $1 billion in 2015. As part of its strategy to diversify its businesses, it invested in a 225 MW thermal power plant that has been operational since 2010. Sapphire Group is owned by Sapphire Textile Mills Limited or STML (55%), Sapphire Fibres Limited (5%), Diamond Fabric Limited (5%), and Sapphire Finishing Mills Limited (5%). STML is listed on the Pakistan Stock Exchange and has a market capitalization of about $131 million.

[7] BAFL, incorporated in 1997, is a publicly listed bank owned and operated by the Abu Dhabi Group, a private equity company owned by Sheikh Mansour bin Zayed El Nahyan, a member of the Abu Dhabi royal family and minister of presidential affairs for the United Arab Emirates. As of 31 March 2016, BAFL had an asset base of $8.2 billion and an equity base of $509 million. In December 2014, the International Finance Corporation acquired a 15% shareholding in BAFL.

2017 and full construction activities commenced thereafter. The final disbursement occurred on 3 April 2018. The project reached its commercial operations date on 14 September 2018, earlier than the date required (November 2018).[8] The project was completed at a cost of $316.8 million, 12% below budget. The Sapphire Group was able to complete the project ahead of schedule and below budgeted costs as a result of (i) lessons learned in establishing its first wind IPP and (ii) working with the same EPC contractor, HydroChina.

16. The first principal debt repayment for the project occurred on 31 March 2019. To date, debt servicing has been timely with no delays or shortfalls.

17. In the first 2 full years of operation (fiscal years 2020 and 2021), the TWPP generated 360 GWh and 438 GWh of energy. Delivered energy was 23% lower than the target of 520 GWh per year in the report and recommendation of the President (RRP). This was partially due to lower-than-expected wind speeds but mainly due to offtaker curtailment. The offtaker was curtailing delivery from the TWPP significantly because of economic factors. When facing oversupply, the offtaker chose to dispatch energy from thermal liquefied natural gas power plants, to avoid the higher “take or pay” costs.

18. Following concerns raised by the Wind IPP Associations and overall higher energy demand, curtailments have drastically declined. Curtailments for the TWPP peaked in fiscal year 2020, reflected by NPMV invoicing of 89.5 GWh (17% of the RRP generation target). For FY2021 NPMV declined to 10.1 GWh (2% of the RRP generation target).

19. Despite lower-than-forecast generation, the project’s revenues have been broadly in line with those identified in the RRP, owing to NPMV compensation for curtailments, under the EPA. Profitability has been higher than identified in the RRP, as the Sapphire Group conducted a resource optimization exercise across all its power projects to reduce operating costs. Yet the project has experienced a growing volume of receivables owing to the power sector’s worsening circular debt issue. Receivables due from the offtaker have been averaging $40 million and reached $40.6 million as of May 2022.

20. In 2020, the government had declared circular debt a national emergency, calling for broad-based sector reforms. It formulated a reform plan that included renegotiating IPP tariffs. In return for agreeing to tariff reductions, the government agreed to settle IPPs’ outstanding receivables.

21. Following agreement to the government’s tariff reduction requests by most thermal IPPs, in 2021 the TWPP also in principle agreed to a tariff reduction and entered into a memorandum of understanding (MOU) with the government. The reduction will decrease the return on equity from 17% to 13% and the tariff’s operation and maintenance component by 20%. In return, the government will pay the TWPP’s outstanding receivables. TBCC also negotiated with the government to improve the NPMV mechanism to compensate for actual lost generation versus the existing flat 35% capacity factor-based formula.

22. Yet, to date, TBCC has not formalized the tariff reduction agreement with the government. In 2021 the government executed tariff amendment agreements with most thermal IPPs and government-owned power plants, covering a majority of the relevant capacity targeted for tariff

[8] Project A achieved completion on 15 August 2018, 20 days ahead of schedule (schedule: 4 September 2018); project B achieved completion on 10 September 2018, 23 days ahead of schedule (schedule: 3 October 2018); and project C achieved completion on 13 September 2018, 52 days ahead of schedule (schedule: 4 November 2018).

rationalization. Although the wind IPP sector had in principle agreed to enter into tariff amendment agreements, the government was slow to advance the matter.

23. Following the April 2022 change in government, the new regime has backed away from pursuing tariff renegotiations. Tariff renegotiations were viewed as the initiative of the previous government and did not align with the new government's pro–private sector policy.

II. EVALUATION

A. Project Rationale and Objectives

24. The project's objectives, as set out in the in the RRP, [9] were (i) ensured access to affordable, reliable, sustainable, energy for all; (ii) improved power supply; and (iii) improved power generation mix through investment by the private sector. The project was intended to address Pakistan's severe power shortages. The project's successful implementation was expected to foster investors' and lenders' confidence and promote further private sector investment in renewable energy.

25. The project's objectives were consistent with (i) ADB's Midterm Review of Strategy 2020,[10] (ii) ADB's country partnership strategy for Pakistan 2015–2019,[11] and (iii) ADB's energy policy underlining investments to maximize access to energy for all and promote renewable energy.[12] The project rationale was also consistent with the government's target of achieving 9,700 MW of renewable power by 2030.[13]

26. ADB's participation in the TWPP also reflected ADB's continued involvement in developing the wind power sector. ADB had taken the lead in structuring and financing of pioneering wind projects, namely Zorlu Enerji and Foundation Wind I and II, given local and international banks' long-term lending constraints and reluctance to finance renewable energy initiatives. The TWPP has contributed to the sector's evolution by being one of the largest wind IPPs with a more competitive tariff than earlier-vintage wind projects.

B. Development Results

1. Contributions to Private Sector Development and ADB's Strategic Development Objectives

27. Contributions to private sector development are rated *satisfactory.* This is based on the project's *satisfactory* contributions to ADB's strategic development objectives and private sector development, and the *satisfactory* achievement of project design and monitoring framework (DMF) indicators (paras. 30–33).[14]

28. The TWPP has supported the strategic priorities of ADB's Midterm Review Strategy 2020, addressing environment and climate change through clean energy and infrastructure development and promoting private sector development by commissioning Pakistan's largest

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[9] ADB. 2016. *Report and Recommendation of the President to the Board of Directors: Proposed Loan Triconboston Consulting Corporation (Private) Limited, Triconboston Wind Power Project (Pakistan)*. Manila.
[10] ADB. 2014. *Midterm Review of Strategy 2020: Meeting the Challenges of a Transforming Asia and Pacific.* Manila.
[11] ADB. 2015. *Country Partnership Strategy: Pakistan, 2015–2019.* Manila.
[12] ADB. 2009. *Energy Policy.* Manila.
[13] Government of Pakistan. 2006. *Policy for Development of Renewable Energy for Power Generation.* Islamabad.
[14] TBCC reported achievements for the DMF indicators on a calendar-year basis.

wind IPP, thereby supporting expansion of the wind power sector and lowering tariffs. The project also aligns with the country partnership strategy for Pakistan 2015–2019.[15]

29. The project has reduced fossil fuel use and supported improvement in the energy mix, promoting energy security by using indigenous wind resources, consistent with ADB's 2009 Energy Policy.[16] At the time of loan processing in 2016, wind power generation capacity was nominal at only 306 MW of installed capacity, or 1.2% of the country's total installed capacity. As of June 2021, total installed wind generation capacity had reached 1,248 MW, or 3.4% of total installed capacity.

30. The DMF outcome "Supply of renewable energy from wind sources increased" was evaluated using the following performance indicators: (i) energy delivered, (ii) greenhouse gas emissions avoided, and (iii) direct employment created during operations. .

31. The actual energy delivered by the TWPP for 2021 was 498 GWh, 4.2% below the DMF outcome target for 2020 of 520 GWh. Greenhouse gas emissions avoided for 2021 were 368,245 tCO_2, 3% below the 2020 target of 380,000 tons of carbon dioxide (tCO_2).[17] Energy delivered, and consequently emissions avoidance, was lower than the DMF target because of both low wind speeds and grid curtailments. Less-than-anticipated wind was an identified risk in the DMF.

32. The DMF outcome for jobs created during operations has been met. The TWPP employed 43 local workers in full-time jobs for operations in 2019, against a target of 35 by 2020.

33. The project's DMF outputs have been met. The project achieved its commercial operation date in September 2018, with an installed capacity of 149 MW, against the target of 150 MW by November 2018. During construction it incurred total costs of $40 million for domestic goods and services, higher than the $31 million target. TBCC reported a total of 1,685 jobs created during construction, of which 352 were full-time employees, higher than the DMF target of 300 full-time employees.

34. The project has also been highly impactful for the development of Pakistan's wind power sector. Its scale, relative to the size of completed wind projects at the time, injected confidence in the sector, spurring growth and attracting new debt and equity investors. The project increased competitive pressures, reducing wind tariffs in comparison with earlier projects, and paved the way for tariff reduction in the next generation of wind IPPs. TBCC also implemented several innovations and new ways of doing business that benefited the sector as a whole. The private sector development indicators and ratings appear in Appendix 2.

2. **Economic Performance**

35. The project's economic performance is rated *satisfactory.*

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[15] The country partnership strategy for Pakistan, 2015–2019, emphasized development of six sectors including energy, infrastructure development to attract private sector development, synergies with ADB's nonsovereign operations to increase the supply of indigenous energy (including wind), and environment and climate change generation of renewable energy.

[16] Support of renewable energy is one of the three pillars of ADB's Energy Policy.

[17] Greenhouse gas emissions avoidance is calculated by multiplying the energy delivered to the offtaker by the grid emission factor of 739.73 tons per GWh, following the formula stated in the RRP. This is higher than the GHG emission reduction calculated in the economic analysis which is based on only 10% non-incremental output. Furthermore, in line with the ex-ante economic analysis, the avoidance calculation used an emission factor of 808 tCO_2/GWh for diesel in Pakistan.

3. Environment, Social, Health, and Safety Performance

36. The project is rated *satisfactory* for environmental, social, health, and safety performance. In compliance with ADB's Safeguard Policy Statement (2009), the project was classified as category B for environment, category C for involuntary resettlement, and category C for indigenous peoples and was processed as a standard project loan. TBCC prepared an integrated initial environmental examination, including a cumulative environmental and social impact assessment, a social due diligence report, and an environmental and social management plan. In accordance with the safeguards requirements, TBCC also completed and submitted four annual environmental and social monitoring reports, covering the period from July 2017 to June 2021. During construction and operation, no significant environmental, health and safety issues, incidents, or fatalities have occurred or have been raised by third parties or government. The project did not require any additional land and did not report any unanticipated impacts. TBCC has confirmed compliance with labor standards regulated by national laws, including core labor standards. TBCC also implements a stakeholder engagement plan which guides the process for continuous engagement throughout the project life cycle.

37. Based on the review and evaluation of available reports and other relevant project documents such as the sub-plans developed under the environmental and social management plan, there are no significant outstanding environmental and social safeguard incompliances, including occupational and community health and safety issues or claims. With a view to improve, TBCC is advised to continue submitting annual monitoring reports to ADB and to provide updates on (i) implementation of the measures to improve the waste management plan, (ii) installation of safety signage and barricades around the remaining excavated lands, (iii) the search for an ornithologist's recommendations on restoring excavated land to prevent birds from congregating, and (iv) assessment and planting of *Commiphora wightti* (guggul), a threatened tree.

4. Business Success

38. The project's financial model was updated to reflect the actual figures for FY2016–2021 and the revised projections for FY2022–2038 on the basis of the renegotiated MOU tariff. To be conservative, the renegotiated tariff has been assumed, although it is not expected that the government will pursue TBCC to rationalize its tariff in line with the MOU.

39. The project is rated *excellent* for business success.

5. Overall Development Results Rating

40. The overall development results rating is *satisfactory* based on the project's satisfactory overall contribution to private sector development and ADB's strategic development objectives, adherence to ADB's environment and social safeguard policies, and the project's robust economic and business fundamentals.

C. ADB Additionality

41. The transaction is the largest wind power project in the country and financed by the private sector. At the appraisal stage, local long-term dollar financing was not readily available. TBCC did explore Pakistani rupee financing; however, restrictions on local banks' appetite meant that the project would need to assemble a syndicate of 10 to 15 banks to raise the $300 million debt, which was deemed impractical or unfeasible by sponsors, especially as most project costs were

denominated in US dollars. ADB played an important role in providing long-term, US dollar financing.

42. Sapphire Textile Mills Limited (STML) had been engaged with ADB since 2010 for developing a wind power project. STML valued ADB's engagement, given ADB's role in the sector's development and knowledge in structuring wind projects.

43. ADB's presence reassured the sponsors to undertake a project of this scale, ensuring the government would adequately implement the newly established upfront tariff regime. The TWPP was the second project from the sponsors that adopt an upfront tariff, one that was lower than the earlier cost-plus tariff and had a higher risk profile as there were no government wind-speed risk guarantees.

D. **ADB Investment Profitability**

44. ADB investment profitability is rated *satisfactory*.

45. The pricing was obtained on the basis of benchmark pricing of similar ADB-approved transactions in Pakistan's power sector.

E. **ADB Work Quality**

46. ADB's effectiveness in screening, appraisal, and structuring is rated *satisfactory*. The project is in line with ADB's country, energy sector, private sector development, and environmental protection strategies. The ADB loan was processed within a reasonable period, which demonstrated ADB's responsiveness to clients' needs without compromising standards for good practice. Following concept clearance on 8 June 2016, ADB completed the financial, technical, environmental, social, and legal due diligence by mid-October 2016. Board approval was obtained on 27 October 2016, and the financing agreements were signed on 10 April 2017.

47. ADB's monitoring and supervision quality is rated *satisfactory*. ADB is up to date and well informed on the project's progress and performance. ADB receives quarterly operational and financial reports, audited financial statements, and environmental and social monitoring reports. The report submissions were complemented by regular communication with the project's management team, as well as review missions and site visits. Within ADB, monitoring reports were prepared, most of them on an annual basis. ADB's environmental and social (E&S) team monitored the social and environmental safeguards vigilantly and provided feedback on the environmental, health, and safety (EHS) monitoring done by the project team.

48. ADB's team diligently tracked inputs, activities, outcomes, and other aspects of the project during the design approval and construction phase by conducting site visits and providing guidelines on the E&S monitoring to improve the project's development impact. TBCC valued ADB's quick processing of loan disbursement requests during construction—especially given the complex escrow structure in place with the EPC contractor— as it enabled timely delivery of equipment and ahead-of-schedule completion of the project.

49. Waiver requests have been addressed in a balanced manner, considering both the project's needs and ADB's interests

50. ADB has also played a key role during the government's tariff renegotiation process. Since 2020 ADB has led a discussion group of private sector development financial institutions (DFIs)

in reviewing and strategizing a unified DFI position with the government and IPPs on the tariff renegotiation process. ADB has also engaged directly with the government on this matter, attempting to ensure the tariff renegotiation process remains as orderly and transparent as possible.

51. Overall, ADB's overall work quality is rated *satisfactory*.

F. Overall Evaluation

52. Overall, the project is rated *successful.* Table 1 shows the overall assessment of the project.

Table 1: Evaluation Summary

Indicator	**Unsatisfactory**	**Less than Satisfactory**	**Satisfactory**	**Excellent**
Development Results			**X**	
(i) Contributions to Private Sector Development and ADB Strategic Development Objectives			X	
(ii) Economic Performance			X	
(iii) Environmental, Social, Health, and Safety Performance			X	
(iv) Business Success				X
ADB Additionality			**X**	
ADB Investment Profitability			**X**	
ADB Work Quality			**X**	
(i) Screening, Appraisal and Structuring			X	
(ii) Monitoring and Supervision			X	
	Unsuccessful	**Less than Successful**	**Successful**	**Highly Successful**
Overall Rating			**X**	

ADB = Asian Development Bank.
Source: Asian Development Bank.

III. ISSUES, LESSONS, AND RECOMMENDED FOLLOW-UP ACTIONS

A. Issues and Lessons

53. **Pioneering projects have a high development impact.** Projects like the TWPP, which are proportionately large and are early adopters (of an untested regulatory regime or upfront tariff) can have substantial impact on private sector development. ADB's involvement attracts pioneering sponsors, and successful project implementation further attracts new debt and equity investors. ADB's involvement in the TWPP has had demonstrable effects and has promoted private sector investment in Pakistan's wind power sector. The significance of this promotion is underscored by the fact that at the time, relevant parties such as local equity investors and domestic banks were reluctant or had limited appetite to invest in wind IPPs. This trend has since changed, as seen in the increased participation of local sponsors and banks in wind IPPs.

54. **Selection of suitable sponsors is critical for success, especially for pioneering projects.** STML's prior experience in the power sector has been instrumental in the successful implementation of the project. This experience enabled the TWPP to be constructed on time and within budgeted costs. It also drove the proactive involvement of the project with the grid operator and offtaker to optimize processes and procedures. These have in turn benefited the wind power sector as a whole.

55. **The catalytic impact of pioneering projects is more substantial than the immediate impacts.** As exhibited by the project's tariff, although generation cost is competitive, the real impact was how drastically generation costs and tariffs have declined for subsequent generations of wind IPPs. Projects such as the TWPP that are early adopters bring high multiplier effects in terms of overall development.

B. Recommended Follow-Up Actions

56. ADB will continue to monitor the project's generation, particularly tracking curtailments and their impact on the project's ability to achieve its targets for generation and greenhouse carbon emissions avoidance. ADB will also continue regular communication with other DFIs as and when there are further developments regarding tariff renegotiations.

PROJECT-RELATED DATA

Table A1.1: Investment Identification

1.	Country	Pakistan
2.	Project Number	50200-001
3.	Loan Number	3448
4.	Type of Business	Renewable energy generation - wind
5.	Project Title	Triconboston Wind Power Project
6.	Borrower	Triconboston Consulting Corporation (Private) Limited
7.	Amount of Approved ADB Assistance	ADB Loan – $75.00 million

ADB = Asian Development Bank.
Source: ADB. 2016. *Report and Recommendation of the President to the Board of Directors on a Proposed Loan to Triconboston Consulting Corporation (Private) Limited for the Triconboston Wind Power Project. Manila.*

Table A1.2: Investment Data

1.	Concept Clearance Approval	8 June 2016
2.	Date of Board Approval	27 October 2016
3.	Signing Date of Legal Agreements	10 April 2017
4.	Date of Loan Effectiveness	10 April 2017
5.	Loan Amount and Date of Initial Disbursement	$75 million; 1 August 2017
6.	Loan Terms	11 years, including a 2-year grace period
7.	Loan Maturity	30 September 2028

ADB = Asian Development Bank.
Source: Asian Development Bank.

Table A1.3: Summary of Project Cost and Funding Sources

Project Item	**RRP Amount** ($ million)	**Actual Amount** ($ million)	**Share of Total** (%)
EPC costs (civil and electromechanical works)	253.0	253.0	79.9
Project development costs (feasibility studies, engineering, advisors, and overheads)	9.0	15.9	5.0
Insurance and Sindh sales tax	6.9	5.0	1.6
Custom duties	2.3	2.6	0.8
Working capital	12.2	3.5	1.1
Debt service reserve account funding	27.2	5.0	1.6
Financing lenders' costs (lenders advisors, fees, and due diligence costs)	14.7	11.1	3.5
Total Base Case Project Costs	**325.3**	**296.0**	**93.4**
Contingencies	23.5	12.0	3.8
Interest during construction	11.2	8.8	2.8
Total Project Cost	**360.0**	**316.8**	**100.0**
Sources of Funds			
A. Equity			
Sapphire Group	63.0	62.2	19.6
Bank Alfalah Limited	27.0	7.3	2.3
Baltoro Growth Fund		9.7	3.1
Subtotal (A)	**90.0**	**79.2**	**25.0**
B. Loans			
Asian Development Bank	75.0	66.0	20.8
Deutsche Investitions- und Entwicklungsgesellschaft	45.0	39.6	12.5
International Finance Corporation	75.0	66.0	20.8
Islamic Development Bank	75.0	66.0	20.8
Subtotal (B)	**270.0**	**237.6**	**75.0**
Total Sources of Funds (A+B)	**360.0**	**316.8**	**100.0**

ADB = Asian Development Bank, RRP = report and recommendation of the President.

Note: Numbers may not sum precisely because of rounding.
Sources: ADB. 2016. *Report and Recommendation of the President to the Board of Directors on a Proposed Loan to Triconboston Consulting Corporation (Private) Limited for the Triconboston Wind Power Project. Manila*; and Triconboston Consulting Corporation (Private) Limited.

Table A1.4: Data on Asian Development Bank Missions

Mission	Dates	Person-Days	Persons	Purpose
Investment appraisal	27–29 June 2016	9	3	Environment and social
Investment appraisal	20–22 July 2016	6	2	Legal
Investment appraisal	14 October 2016	1	1	Legal
Investment appraisal	21–23 November 2016	12	4	Negotiation
Project administration	29–30 January 2018	4	2	Investment administration

Source: Asian Development Bank.

RESULTS AND RATINGS FOR PROJECT CONTRIBUTIONS TO PRIVATE SECTOR DEVELOPMENT AND ADB STRATEGIC DEVELOPMENT OBJECTIVES—INFRASTRUCTURE

Results Area	Actual Achievements	Rating[a]	Justification	Potential Future Achievements	Risk
1. Within-company PSD effects					
1.1 **Improved skills.** New or strengthened strategic, managerial, operational, technical, or financial skills	Domain knowledge on wind IPP construction and operations was strengthened.	Excellent	At the start of construction, key skilled O&M operator and EPC contractor staff were foreigners. Over the past 5 years, local capabilities have been developed and most O&M operator staff and EPC contractor staff, including managers, are local hires.	Continued growth and skill development of local staff for construction and O&M of wind power plants	Low
1.2 **Improved business operations.** Improved ways to operate the business and compete, as seen in investee operational performance against relevant best industry benchmarks or standards	The project established NPMV procedures with the offtaker for FiT wind IPPs.	Excellent	TBCC prepared operating procedures that included standard operating procedures for forecasting and NPMV compensation.	Establishment of procedures, setting precedents that will continue to be used for future wind projects	Low
1.3 **Improved governance.** As evident in standards set related to corporate governance, stakeholder relations, EHS fields, and/or energy conservation, and their implementation	The project achieved excellence in E&S standards.	Excellent	The TWPP is the only wind project in Pakistan to have been rated "excellent" for E&S compliance with IFC Performance Standards PS1, PS2, PS3, and PS4.	Implementation of adequate E&S standards in future projects of TWPP sponsors	Low
1.4 **Innovation**. New or improved infrastructure design, technology, service delivery, or ways to cover or contain costs, manage demand, or optimize utilization; improved risk allocation between private companies and government, financial structure, etc.	The project implemented technical, design and process innovations benefiting other wind IPPs and grid operators and offtakers.	Excellent	The TWPP is the first project in Pakistan to install vortex generators on its turbines, helping improve energy yields. Subsequently, other wind projects have used these generators. The TWPP was also the first project to use an oversized main transformer design, to ensure	Application of innovation and lessons of the project, continuing to benefit future projects established in Jhimpir	Low

Results Area	Actual Achievements	Rating[a]	Justification	Potential Future Achievements	Risk
			full evacuation capacity and 100% redundancy. The TBCC team made design changes to the WTG doors to address dust infiltration issues (inherent in Jhimpir). This solution was later adopted by the operator for all wind projects in Jhimpir. The TWPP was the first wind project to formally submit energy forecasts to the power purchaser and grid operator, to facilitate the grid operator's planning and reduce curtailments. The project is the only wind project in Pakistan to have conducted a power curve test, which confirmed the O&M contractor's power curve guarantee.		
1.5 **Catalytic element.** Mobilizing or inducing more local or foreign market financing or foreign direct investment in the company	The project mobilized foreign debt and local equity investment.	Satisfactory	In addition to Sapphire Group, the project attracted equity investment from first-time investors in the sector Bank Al-Falah and the Baltoro Growth Fund. Moreover, the TWPP was DEG's first investment in Pakistan's power sector. Since then, DEG has provided debt to three other wind IPPs.	Continued investment in the sector by the TWPP's equity investors and by DFI lenders in Pakistan's renewable power sector	Medium Appetite for investment in the debt and equity power sector constrained by country macroeconomic concerns and power sector issues, i.e., circular debt, tariff renegotiations
2. Beyond company PSD effects					
2.1 **Private sector expansion.** Contribution by	The TWPP has facilitated the sector's	Excellent	The construction of the largest wind power project infused	Continued expansion of the wind power sector,	Medium

Results Area	Actual Achievements	Rating[a]	Justification	Potential Future Achievements	Risk
a pioneering or low-profile project that facilitates in its own right, or paves the way for, more private participation in the sector and economy at large	development and expansion.		investor confidence in the sector, enabling development of further projects. As of June 2016 (time of RRP processing), Pakistan had six wind IPPs, with a total installed capacity of 306 MW, or 1.2% of total installed capacity. As of 2022, there are 26 wind IPPs with a total installed capacity of 1,335 MW, ~3.4% of total installed capacity.	resulting in greater private sector involvement and an increased share of Pakistan's total installed generation capacity 10 wind IPPs, totaling 510 MW of installed capacity, under construction[b]	
2.2 **Competition**. Contribution of new competition pressure on public and/or other sector players to raise efficiency and improve access and service levels in the industry	The TWPP set a new sector benchmark, driving more competitive tariff structures.	Excellent	The project has paved the way for lower tariffs. Its FiT tariff was 24% lower than the average reference tariff for all previous wind projects ($0.1045/kWh vs. $0.1382/kWh).	More competitive cost structure in wind IPPs under construction, with the average reference tariff being 55% lower than the TWPP's tariff ($0.0474/kWh vs. $0.1045/kWh)	Low
2.3 **Demonstration effects**. Adoption of new skills, improved infrastructure assets and services, more efficient processes and maintenance regimes, improved standards, risk allocation, and mitigation beyond the project company	The TWPP's procedures, innovations and learnings are being applied across the sector, enhancing performance.	Excellent	The TWPP's energy forecasting and NPMV compensation operating procedures have been adopted by the grid operator and other wind projects. Moreover, the energy forecast has assisted the grid operator and offtaker in addressing bottlenecks and improved dispatch, lowing curtailment. The TWPP also continues to work with the grid operator and conducts technical analysis to periodically check the overall grid health. This has improved grid reliability and availability Other wind IPPs have also used vortex generators in their WTGs. Moreover, to address dust infiltration the EPC contractor	Continued application of experience from the TWPP by sector players, benefiting all new projects developed in the Jhimpir corridor	Low

Results Area	Actual Achievements	Rating[a]	Justification	Potential Future Achievements	Risk
			(HydroChina) has applied the same WTG door design in 10 other projects it has developed in Jhimpir.		
2.4 **Linkages**. Relative to investments, the project contributes notable upstream or downstream linkage effects to business clients, consumers, suppliers, key industries etc. in support of growth.	The project developed local capacity to build and operate wind power projects in Pakistan.	Satisfactory	Increased proportion of local skilled labor and expertise.	Continued development of local skills and capabilities	Low
2.5 **Catalytic element.** Mobilizing or inducing more local or foreign market financing or foreign direct investment in the sector (beyond the company) through pioneering or catalytic finance.	The project catalyzed debt and equity financing for wind IPPs.	Excellent	Twelve wind IPPs achieved financial close in 2019. They had a total capacity of 600 MW and a total cost of approximately $750 million.[c] These projects were implemented by various local sponsors, the first wind power investment for several of them. Sponsors included Lucky Group, Metro, Yunus Brothers, Gul Ahmed, and Liberty Mills. In terms of DFI involvement, the IFC financed 6 of the projects, the DEG expanded its exposure to 3 projects, and the CDC financed 3 projects. Earlier, local banks had limited exposure to wind IPPs. All 12 projects were financed by local banks. Important to highlight: in addition to the large local banks, they included second-tier ones such as Faysal Bank, Bank Alfalah, and the Bank of Punjab.	Continued attraction of private sector debt and equity investment to wind IPPs, furthering sector growth	Medium Appetite of debt and equity investors for the power sector constrained by country macroeconomic concerns and power sector issues, i.e., circular debt, tariff renegotiations
2.6. **Affected laws, frameworks, regulation.**	Cost tariff structures were implemented.	Satisfactory	The success of the TWPP and other FiT wind IPPs enabled	Continued rationalization of wind energy tariffs	Medium

Results Area	Actual Achievements	Rating[a]	Justification	Potential Future Achievements	Risk
Contributes to improved laws and sector regulation for PPPs, concessions, joint ventures, and build–operate–transfer projects, and liberalizing markets as applicable for improved sector efficiency.			the government to obtain bids for more cost-effective, less expensive wind IPP tariffs (please refer to section 2.2).		
3. Contribution to other ADB strategic objectives					
3.1 **Sector development outputs.** Contribution to other sector development outputs and outcomes not captured under point 2, such as capacity or network expansion	The proportion of private sector participation in Pakistan's energy sector has increased.	Satisfactory	The share of the private sector in power generation capacity has increased to 51%.	Further growth in private sector participation	Medium Low benchmark tariffs for auctions of renewable energy projects
3.2 **Sector development outcomes.** Contribution to other sector development outputs and outcomes not captured under point 2, such as increased infrastructure utilization or consumption, improved in-country connectivity, and improved energy security	Energy security has improved through the increased use of indigenous and sustainable resources such as wind and solar.	Satisfactory	The project is contributing significantly to achieving Pakistan's renewable energy goal of 30% by 2030.	Continued growth of the renewable sector, as evidenced by the NTDC's Integrated Generation Capacity Expansion Plan, which will further strengthen energy security	Medium Low benchmark tariffs for competitive bidding of renewable energy projects
3.3 **Inclusion.** Improved access to, availability or affordability of infrastructure services for the poor and other disadvantaged groups	The project has improved access to roads, schools, and medical facilities for the local community.	Satisfactory	In line with TBCC's CSR objectives, the project is working to improve the standard of living for local communities.	TBCC continuing the project's CSR initiatives to ensure continued growth of local communities	Low
3.4 **Job creation**. Creation of additional sustainable jobs or self-employment; distinguish between jobs created within and beyond the company	The TWPP generated employment opportunities for locals from the community in Jhimpir during the	Satisfactory	The TWPP employed 1,685 people during construction and about 40 during commercial operations.	Continued employment opportunities for locals	Low

Results Area	Actual Achievements	Rating[a]	Justification	Potential Future Achievements	Risk
	construction and operation phases.				
3.5 **Environmental sustainability.** Project net impact on GHG emissions; any other contributions to environmental improvements	The project is generating clean energy and playing an important role in reducing GHG emissions.	Satisfactory	On average, the projects avoided greenhouse gas emissions of 297,765 tCO_2/year (based on data for FY2019–2020 and FY2020–2021).	Continued offsetting of carbon emissions	Low
3.6 **Regional integration.** Project contributions to regional cooperation and integration by facilitating trade, cross-border mobility, cross-border power supplies, etc.	The project used overseas companies for construction and operations.	Satisfactory	HydroChina was the project EPC contractor and O&M operator for the first 2 years of operation.	Pursuit of more EPC contracts by HydroChina in 10 other wind IPP projects in Pakistan	Low
4. Overall Rating[b]		**Satisfactory**			

ADB = Asian Development Bank; CDC = Commonwealth Development Corporation, CSR = corporate social responsibility; DEG = Deutsche Investitions- und Entwicklungsgesellschaft; DFI = development financial institution; E&S = environmental and social; EPC = engineering, procurement, and construction; FiT = feed-in tariff; FY = fiscal year; GHG = greenhouse gas; GWh = gigawatt-hour; IFC = International Finance Corporation; IPP = independent power producer; MW = megawatt; NPMV = non-project missed volumes; NTDC = National Transmission and Despatch Company; O&M = operation and maintenance; PS = performance standard; PSD = private sector development; PPP = public–private partnership; RRP = report and recommendation of the President; SWPCL = Sapphire Wind Power Company Limited; tCO_2 = tons of carbon dioxide; WTG = wind turbine generator.

[a] The rating scale for each results area is Unsatisfactory, Less than Satisfactory, Satisfactory, Excellent, Not applicable.

[b] Alternate Energy Development Board

[c] NEPRA State of the Industry Report 2021.

[d] The overall rating scale is Unsatisfactory, Less than Satisfactory, Satisfactory, Excellent.

Source: Asian Development Bank.

SECTOR REVIEW

A. Overview of Pakistan's Energy Sector

1. The Power Division of the Federal Ministry of Energy is responsible for providing strategic guidance, setting policies, and coordinating and supervising organizations in the power sector. The provincial governments also have energy departments that can and do implement power projects within provinces. The electricity sector is regulated by the National Electric Power and Regulatory Authority (NEPRA), whose independent mandate oversees generation, transmission, and distribution. Power generation is provided by state-owned generation companies for thermal power, the Water and Power Development Authority for hydropower, and independent power producers (IPPs) for other kinds of power.

2. The Government of Pakistan opened the power sector to private investment in its 1994 Power Policy. It created the Private Power and Infrastructure Board to implement the policy and to facilitate investment through a single-window operation. The policy was followed by the 1995 Hydroelectric Policy, which was designed to encourage investment in hydropower generation. The subsequent 2002 Power Policy covered concessions for development of all private sector thermal and hydro generation plants producing more than 50 megawatts (MW), leaving projects of less than 50 MW to provincial governments. In 2006, the government announced the Renewable Energy Power Policy, intended to encourage energy development from renewable sources with a goal to achieve 9,700 MW of renewable power in the energy mix by 2030.

3. The national electricity transmission system is run by the state-owned National Transmission & Despatch Company (NTDC), which transmits power between generators and the state-owned distribution companies that serve consumers. The NEPRA Amendment Act, 2018 completely changed the concept of distribution companies having exclusive rights to distribute electricity to consumers in their respective service territories. NEPRA, in its earlier reports, has emphasized that the majority of power is purchased from generators on behalf of distribution companies by the Central Power Purchasing Agency, which acts as the power market operator. The exception is K-Electric Limited, a vertically integrated, private company that provides power generation, transmission, and distribution for the Karachi metropolitan area. The Alternative Energy Development Board, an autonomous body under the Ministry of Energy, promotes and implements renewable energy projects. That role is critically important, given the need to use domestic natural resources to curtail imports of fossil fuel.[1]

4. Other agencies in the power sector include the Planning Commission, which carries out integrated energy planning; the Private Power and Infrastructure Board, which functions as a "single-window" office for private developers in power generation; and the National Energy Efficiency and Conservation Agency, which leads energy efficiency and conservation activities through development plans and policies, labeling, information outreach, and training.[2]

B. Energy Generation

5. The power industry in Pakistan has a mix of thermal, hydro, and nuclear power plants. As on 30 June 2021, the share of the installed capacity of power plants of intermittent nature (hydro, wind, solar, and bagasse) in the generation mix was about 30% with 12,062 MW, while the share of thermal power plants, including nuclear power plants, was about 70% with 27,711 MW.

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[1] Institute for Energy Economics and Financial Analysis. December 2018. *Pakistan's Power Future*. Lakewood.
[2] ADB. 2019. *Pakistan: ADB's Support to Pakistan Energy Sector (2005–2017).* Manila.

6. Renewables (hydro, wind, solar, and bagasse) make up 12,062 MW or about 30% of installed capacity as on 30 June 2021. This has increased significantly from 7,424 MW in 2015.

7. The total installed generation capacity of public sector power plants in the country was 20,820 MW (52%) as of 30 June 2021, and the installed generation capacity of private sector power plants was 18,952 MW (48%). The electricity generated by public sector power plants was 75,875 gigawatt-hours (GWh) (53%), while private sector power plants generated 67,215 GWh (47%).

Table A4.1: Pakistan's Electricity Generation Composition

	June 2021 Capacity		**FY2020–2021 Generation**		**June 2015 Capacity**	
Source	MW	%	GWh	%	MW	%
Thermal	25,098	63	85,914	60	16,814	67
Hydro	9,915	25	40,985	29	7,116	28
Nuclear	2,612	7	11,090	8	787	3
Wind	1,248	3	3,479	2	308	1
Solar	530	1	912	1		-
Bagasse/biomass	369	1	711	0		-
Total	**39,772**	**100**	143,091	**100**	**25,025**	**100**

GWh = gigawatt-hour, MW = megawatt.
Note: Numbers may not sum precisely because of rounding.
Sources: National Electric Power Regulatory Authority. 2016. *State of Industry Report 2015*. Islamabad; and National Electric Power Regulatory Authority. 2021. *State of Industry Report 2021*. Islamabad.

8. To date, all wind-power development has occurred in the Gharo-Keti Bandar corridor in the south of the country, which has a theoretical potential of about 50 GW of wind energy. This corridor combines good wind resources with relative proximity to load centers and national grid connectivity. Pakistan also has good solar radiation resources, particularly in the south and southwestern parts of the country. Areas of high solar radiation include the Thar Desert, parts of which have been allocated for coal mining to supply fuel to some of the proposed coal-fired power plants. In southeast Pakistan, Sindh Province has initiated the Sindh Solar Energy program, which will see utility-scale, distributed, and residential solar installed, including up to 400 MW of capacity in solar parks.

9. Over the years, Pakistan has gone through cycles of either deficits in power generation capacity or undesired surpluses. After facing a long period of capacity shortfall, the country is now facing the issue of underutilization of available capacity. During FY2020–2021, the utilization of power plants of intermittent nature varied between about 15% and 67% of their dependable capacity. The overall utilization of all such power plants remained at about 41% of their combined dependable capacity. During FY2020–2021, the utilization factor of combined base-load thermal power plants in the country remained about 45% of their dependable generation capacity.

10. From 2017 to 2021, the NTDC and K-Electric have not been able to meet demand at peak times, except for the NTDC in 2020. Data projections show that they will be able to meet demand at peak times.

Table A4.2: Power Demand and Supply at Peak Times

FY	Planned Generation Capability (MW) NTDC	KE	Projected Demand During Peak Hours (MW) NTDC	KE	Surplus (Deficit) (MW) NTDC	KE
2017	19,020	2,920	25,117	3,270	(6,097)	(350)
2018	23,766	3,008	26,741	3,527	(2,975)	(519)
2019	24,565	3,196	25,627	3,530	(1,062)	(334)
2020	27,780	3,202	26,252	3,604	1,528	(402)
2021	27,819	3,424	28,253	3,604	(434)	(180)
Projected Figures	**Planned Generation Capability** (MW)		**Projected Demand During Peak Hours** (MW)		**Surplus (Deficit)** (MW)	
FY	**NTDC**	**KE**	**NTDC**	**KE**	**NTDC**	**KE**
2022	32,989	4,317	30,921	4,110	2,068	207
2023	35,896	4,737	31,953	4,422	3,943	315
2024	37,918	3,878	33,696	4,588	4,222	(710)
2025	39,157	5,002	35,422	4,759	3,735	243
2026	42,075	5,002	36,206	4,935	5,869	67

KE = K-Electric Limited, MW = megawatt, NTDC = National Transmission and Despatch Company Limited.
Note: Numbers may not sum precisely because of rounding.
Source: National Electric Power Regulatory Authority. 2021. *State of Industry Report 2021*. Islamabad.

11. Households have been the largest consumers of electricity, followed by industrial, agricultural, and commercial consumers.

Table A4.3: Electricity Consumption by Sector

Consumer Category	Consumption, FY2021 GWh	Consumption, FY2021 Percentage	Consumption, FY2017 GWh	Consumption, FY2017 Percentage
Household	57,856	48	48,055.1	48
Commercial	8,397	7	7,769.6	8
Industrial	29,886	25	23,951.6	24
Agricultural	10,237	8	9,222.1	9
Others	14,830	12	10,610.8	11
Total	**121,206**	**100**	**99,609.2**	**100**

GWh = gigawatt-hour.
Sources: Government of Pakistan. 2017. *NEPRA State of Industry Report*. Islamabad; National Electric Power Regulatory Authority. 2021. *State of Industry Report 2021*. Islamabad.

12. The number of wind IPPs in Pakistan has increased over time with the downtrend tariff scheme, as shown in Table A4.4.

Table A4.4: Wind Energy Generation

Independent Power Producer	Location	Capacity (MW)	Financial Close Date	Commercial Operations Date	EPA Term (years)	Tariff Scheme	Levelized Tariff ($/kWh)
FFC Energy Ltd.	Jhimpir	49.5	28-Jun-2011	16-May-2013	20	Cost plus	0.161090
Zorlu Enerji Pakistan Ltd.	Jhimpir	56.4	29-May-2012	26-Jul-2013	20	Cost plus	0.133456
Three Gorges First Wind Farm Pakistan (Pvt.) Ltd.	Jhimpir	49.5	17-Jul-2013	25-Nov-2014	20	Cost plus	0.139393
Foundation Wind Energy II (Pvt.) Ltd.	Gharo	50.0	3-Apr-2013	10-Dec-2014	20	Cost plus	0.141164

Independent Power Producer	Location	Capacity (MW)	Financial Close Date	Commercial Operations Date	EPA Term (years)	Tariff Scheme	Levelized Tariff ($/kWh)
Foundation Wind Energy I Ltd.	Gharo	50.0	18-Jul-2013	11-Apr-2015	20	Cost plus	0.141359
Sapphire Wind Power Company Ltd.	Jhimpir	52.8	7-Jul-2014	22-Nov-2015	20	Upfront	0.135244
Metro Power Company Ltd.	Jhimpir	50.0	20-Feb-2015	16-Sep-2016	20	Cost plus	0.145236
Yunus Energy Ltd.	Jhimpir	50.0	9-Feb-2015	16-Sep-2016	20	Upfront	0.151278[a]
Master Wind Energy Pvt. Ltd.	Jhimpir	52.8	9-Feb-2015	7-Oct-2016	20	Upfront	0.151088
Act Wind (Pvt.) Ltd.	Jhimpir	30.0	27-Mar-2015	18-Oct-2016	20	Upfront	0.151278[a]
Gul Ahmed Wind Power Ltd.	Jhimpir	50.0	27-Mar-2015	18-Oct-2016	20	Upfront	0.151088
Tenega Generasi Ltd.	Gharo	49.5	27-Mar-2015	11-Oct-2016	20	Upfront	0.143166
Hydro China Dawood Power Pvt. Ltd.	Gharo	49.5	27-Mar-2015	5-Apr-2017	20	Upfront	0.135244
Sachal Energy Development Pvt. Ltd.	Jhimpir	49.5	17-Dec-2015	18-Apr-2017	20	Cost plus	0.141681
UEP Wind Power Pvt. Ltd.	Jhimpir	99.0	30-Mar-2015	16-Jun-2017	20	Upfront	0.135244
Jhampir Power (Pvt.) Ltd.	Jhimpir	50.0	8-Aug-2016	16-Mar-2018	20	Upfront	0.104481
Hawa Energy Pvt. Ltd.	Jhimpir	49.6	8-Aug-2016	15-Mar-2018	20	Upfront	0.104481
Artistic Energy (Pvt.) Ltd. Alternative Energy Pvt. Ltd.	Jhimpir	49.3	26-Dec-2016	16-Mar-2018	20	Upfront	0.121302[b]
Three Gorges Pakistan Second Wind Farm Pakistan Ltd.	Jhimpir	49.5	17-Mar-2017	30-Jun-2018	20	Upfront	0.104481
Three Gorges Pakistan Third Wind Farm Pakistan (Pvt.) Ltd.	Jhimpir	49.5	17-Mar-2017	9-Jul-2018	20	Upfront	0.104481
Tricon Boston Consulting Corporation Pvt. Ltd. - A	Jhimpir	49.6	10-May-2017	16-Aug-2018	20	Upfront	0.104481
Tricon Boston Consulting Corporation Pvt. Ltd. - B	Jhimpir	49.6	10-May-2017	14 spe 2018	20	Upfront	0.104481
Tricon Boston Consulting Corporation Pvt. Ltd. - C	Jhimpir	49.6	10-May-2017	11-Sep-2018	20	Upfront	0.104481
Zephyr Power Pvt. Ltd.	Gharo	50.0	10-May-2017	27-Mar-2019	20	Upfront	0.114629
Lakeside Energy Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047154
Artistic Wind Power Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047212
Liberty Wind Power 1 Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047824

Independent Power Producer	Location	Capacity (MW)	Financial Close Date	Commercial Operations Date	EPA Term (years)	Tariff Scheme	Levelized Tariff ($/kWh)
Indus Wind Energy Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047931
Master Green Energy Ltd.	Jamshoro	50.0	19-Aug-2019	20-Aug-2021	25	Cost plus	0.047227
ACT2 Wind Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	17-Feb-2022	25	Cost plus	0.047212
Liberty Wind Power 2 Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047824
Metro Wind Power Ltd.	Jhimpir	60.0	18-Nov-2019	Under construction	25	Cost plus	0.046360
Nasda Green Energy Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047190
DIN Energy Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047824
Gul Ahmed Electric Ltd.	Jhimpir	50.0	18-Nov-2019	Under construction	25	Cost plus	0.047212
Tricom Wind Power Pvt. Ltd.	Jhimpir	50.0	18-Nov-2019	16-Feb-2022	25	Cost plus	0.047824

EPA = energy purchase agreement; kWh = kilowatt-hour; MW = megawatt.
[a] PRs107.7/$1 as of 21 November 2013.
[b] PRs10.75/$1 as of 31 March 2016.Sources: National Electric Power Regulatory Authority and Asian Development Bank.

C. Circular Debt

13. The biggest challenge faced by Pakistan's power sector is circular debt, caused by multiple factors. Circular debt is defined as the cash shortfall between revenues and generation costs in the power sector, resulting in liquidity constraints across the energy supply chain. As of 30 June 2021, circular debt had accumulated to PRs2.28 trillion ($14.4 billion), approximately 5% of gross domestic product, and had become one of the country's major economic challenges.

14. The underlying cause of circular debt is exacerbated by issues such as inadequate policy implementation and planning, inappropriate regulation, weak governance, and underinvestment. These factors in turn result in high generation costs, which in turn have not been reflected in end-consumer tariffs.

15. The high costs are in part due to the high operating leverage in the power sector, as approximately 50% of generation costs are fixed-capacity payments (due to take-or-pay power purchase agreements). The sector also has high aggregate technical and commercial losses of ~30% (transmission and distribution losses: 20%, under-recovery losses: 10%). Coupled with an oversupply scenario (with only 50–60% of sector capacity utilized), generation costs are substantially higher than revenues for energy consumed.

16. End-consumer tariffs have been increasing, though not in line with rising generation costs, and are well below cost-recovery levels. To meet the shortfall, the government has paid approximately $7 billion in subsidies over the past 5 years. Subsidies paid have been less than what had been budgeted, not to mention what was required. Pakistan already has some of the highest tariffs among its regional peers, and over the past few years ~47% of industrial consumers have moved off-grid.

17. To address circular debt, the government has been periodically raising debt on its balance sheet (through a government special purpose vehicle, Power Holdings Pakistan Ltd) to inject

liquidity into the sector. To date, it has raised ~$6 billion in debt; however, accumulated arrears in the sector total ~$9 billion. This equals roughly half of total annual generation costs.

18. Circular debt has grown more rapidly over the past 5 years in particular because of the ~40% increase in installed capacity, coupled with a ~40% depreciation in the dollar–rupee exchange rate—important because generation tariffs are indexed to the US dollar.

ENVIRONMENTAL IMPACT

A. Introduction

1. The Triconboston Wind Power Project (TWPP) comprises the design, construction, and operation and maintenance (O&M) of three 50-megawatt (MW) wind power plants (Tricon-A, -B, and -C) and associated infrastructure located in Jhimpir, Thata District, Sindh Province, about 100 kilometers (km) northeast of Karachi. The three wind farms are constructed on three adjacent parcels of land totaling 3,852 acres, allocated and leased by the Government of Sindh. Each wind farm has 1.7 MW General Electric wind turbine generators (WTGs) with a 103-meter (m) rotor diameter and an 80 m hub height. Other project components include a substation for each wind farm, an underground electrical collection system connected to the substation, an operations and control building that houses the substation and grid connection, a network of project roads, a standby generator, a plant O&M facility, and two meteorological masts. The National Transmission and Dispatch Company is responsible for building, operating, and maintaining the 7 km transmission line that evacuates power to the main grid.[1] Triconboston Consulting Corporation Private Limited (TBCC) selected HydroChina as the engineering, procurement, and construction (EPC) contractor and General Electric as the O&M contractor for the project. Construction started in August 2017, and operation commenced in August 2018 for the Tricon-A wind farm and in September 2018 for the Tricon-B and Tricon-C wind farms.

2. The project is classified as environment category B. It is in the Jhimpir wind corridor, which is a rural area with little agricultural land owing to the scarcity of water and the arid conditions. The area is generally flat and has wind conditions favorable to the development of wind energy power plants. The project site is 20 km from Keenjhar Lake, also known as Kalri Lake, which is included in the Ramsar list of internationally important wetlands. The site is not along a major fly route of migratory birds. The site is not located in other environmentally sensitive areas or areas of cultural significance. No industrial or residential activity is permitted on the land allocated for wind energy projects. Initial environmental examination (IEE) reports for each wind farm were prepared and submitted to the Sindh Environmental Protection Agency, and a No Objection Certificate was issued for all three in January 2017. Because of the increasing number of wind farms constructed along the corridor, a separate integrated initial environmental examination (IIEE), including a cumulative environmental and social impact assessment, was prepared in June 2016 as part of the due diligence to comply with the requirements of ADB and the International Finance Corporation. An environmental and social management plan (ESMP) outlining the key issues and concerns and the measures to mitigate impacts during construction and operation was included in both the IEE and the IIEE.

3. The extended annual review has evaluated the status of implementation of the ESMP and reviewed relevant environmental and health, safety, and environmental (HSE) documents provided by TBCC to gauge compliance of the project with the ADB Safeguard Policy Statement (2009), national regulations, and international standards. No physical site visit has been carried out because of the pandemic situation but meetings with the company’s HSE team have been conducted to collect relevant technical and environmental and social (E&S) data and information.

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[1] The National Transmission and Despatch Company is responsible for country-wide transmission of electricity. It has prepared an environmental and social impact assessment and a resettlement plan for the 220-kilovolt Jhimpir grid station and associated transmission lines project and obtained necessary government permits for construction and operation of the facility.

B. Review Findings

4. **Compliance with ADB safeguards requirements and national and environmental regulations.** TBCC developed an IIEE that includes an ESMP. The status of ESMP implementation has been documented and presented in annual monitoring reports, in compliance with ADB Safeguard Policy Statement (2009) requirements. The project complies fully with applicable national laws and regulations for pollution prevention (air, water, noise, waste, soil), occupational health and safety (OSH), hazardous waste management, cultural chance finds, and wildlife protection. Required government permits and clearances have been obtained and updated. No fines or violations have been imposed by the state. Conditions in the No Objection Certificate issued by the Sindh Environmental Protection Agency have been complied with. They include engaging an independent monitoring consultant, which will monitor the compliance of the project and submit quarterly monitoring reports.[2] The TWPP has a working grievance redress mechanism to receive and facilitate resolution of internal and external grievances. All grievances have been closed out. TBCC also developed an E&S management system based on ISO 14001:2015 (Environmental Management System) standards and has conducted training of staff and contractors on ISO 9001 (Quality), ISO 14001 (Environment), and ISO 45001 (OHSAS). TBCC is preparing to resume the ISO certification process, following delays caused by the pandemic. The project also adhered to international standards such as the Environmental, Health, and Safety General Guidelines and Guidelines for Wind Energy (August 2015) of the International Finance Corporation (IFC).

5. **Environmental Management and Monitoring Plans.** TBCC, O&M contractors, and subcontractors are jointly responsible for implementing all measures included in the ESMP. The TBCC HSE team is responsible for system documentation, risk assessment, and conduct of HSE-relevant training, as well as audit, inspection, and investigation of HSE-related activities. The team coordinates with the IMC, which serves as the third-party consultant overseeing implementation of the ESMP. ESMPs for the construction and operational phases have been developed and implemented to manage and mitigate project impacts.[3] Additional management and monitoring plans have been implemented during the operational phase for shadow flicker, birds, and flora and fauna.

6. Construction phase impacts such as generation of noise, vibration, dust, and solid, liquid, and hazardous waste have been adequately managed. Restoration and replantation to compensate for loss of flora during construction of access roads and WTG foundations have been completed. The workers accommodation camp has been retained for further use by the EPC contractor for other projects. Two excavated lands, one within the project boundary and another outside it, have been left open at the request of villagers for use as rainwater storage ponds to manage water requirements for agricultural purposes. The company will provide safety signage and barricades around the excavated lands. The excavated lands are regularly inspected as part of the Erosion and Sedimentation Control Plan, together with access roads, water causeways, and culverts. Recommendations from an ornithologist should be sought if the excavated lands should be restored to original condition to prevent bird congregation. TBCC also installed a 100

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[2] Arch Associates, the selected IMC, serves as a third-party consultant ensuring the compliance of TBCC and the O&M contractor with the ESMP, SEPA conditions, and other national and international standards by conducting audits and inspections, reviewing procedures and policies, and providing guidance on environmental, OHS, and social compliance management and mitigation measures.

[3] This includes standalone monitoring plans for management of air emissions, noise and vibration, ecology, waste, water, hazardous materials, raw materials, security, cultural heritage, and community impact, as well as for erosion and sedimentation control, spill prevention and response, and emergency preparedness and response.

KW solar plant during construction as a cost-saving initiative. The IFC awarded TBCC an "excellent" rating in 2017 for E&S compliance during the construction phase.

7. Operational phase monitoring of impacts on air quality, noise, and shadow flicker showed that results are within acceptable limits. No negative impact of shadow flicker was observed, and no complaints were reported by people living in the temporary settlements used for agricultural and grazing purposes located 100–300 m from the project site. All waste is collected from sources, segregated, stored, and sold to a waste disposal company. Because of pandemic restrictions, the waste management plan has not been properly implemented and waste management has been a challenge. TBCC is committed to carrying out the following measures to improve implementation of the solid and liquid waste management plans: (i) audit of management systems (for solid and liquid wastes) of the waste management company, (ii) on-site verification of disposal activities at approved disposal sites, and (iii) monitoring of solid and wastewater collection and disposal activities by the IMC. The company will continue to provide annual monitoring reports to ADB. Updates on measures to improve the waste management plans will be reported in the next annual monitoring report submitted to ADB.

8. **Occupational Health and Safety Plan.** An OHS plan was developed and integrated into the project's O&M activities. It consists of job hazard analysis and system procedures for OHS risk assessment, inspection and audit, incident and accident reporting and investigation, and special work procedures. Daily and weekly inspections are conducted. An emergency preparedness procedure has been developed, and regular training is conducted on electrical safety, fire safety and firefighting, and evacuation procedures. During the construction and operation phases no fatalities have been recorded.

9. **Biodiversity Monitoring and Cumulative Impacts.** Collision risk impacts to birds and bats are considered low because of the types of bird species observed in the area. A program to monitor bird flight activity and collision is incorporated in the ecological management plan and the bird and bat monitoring plan. Weekly bird monitoring (species, count, flight height) and bird and bat carcass removal has been conducted as per the plan. No significant bird collisions or mortalities have been recorded. In addition, the company has implemented measures to minimize operational bird fatalities which includes setting a 200 m distance between WTGs, use of underground electrical lines from WTGs to substations, use of flash lamps on WTGs at night, and checking of tower holes to reduce nesting. Bird mortality is monitored to assess the exact impact of the project. TBCC has retained a consultant for ornithological studies and bird monitoring, according to the bird and bat monitoring plan, and consults an IFC biodiversity specialist. The company has also conducted vehicle speed control (to prevent dust generation and killing of fauna), regular site inspections, and biodiversity inspections for *Commiphora wighitti* (guggul), in accord with the flora and fauna management plan. Owing to the number of wind farm projects in the Jhimpir region, the IFC has developed a Joint Management and Monitoring Framework for Cumulative Impacts, initiated in 2020 and completed in October 2021.[4] The flora and fauna management plan has been updated to consider the recommendations of the study, which include biodiversity assessment and development of adaptive measures for guggul, following reports of the threatened species in the region. Updates on the assessment and planting of guggul will be reported in the annual monitoring report submitted to ADB.

[4] The Joint Management and Monitoring Framework for Cumulative Impacts initially focuses on the seven proposed wind power projects, namely ACT II, Artistic, Din Energy, Gul Ahmed, Lakeside, Metro 2, and Tricom (referred to as Super 6 and Lakeside), but it can be expanded to other developers and operators in the Jhimpir Wind Farm Region.

C. Conclusion and Recommendation

10. The projects have complied with the E&S requirements in the ADB Safeguard Policy Statement (2009), applicable national laws and regulations, and relevant international standards. This conclusion is based on the review and evaluation of relevant E&S documents (such as the IEE, the IIEE, and the cumulative environmental and social impact assessment, the ESMP, the Joint Management and Monitoring Framework, and annual environmental and social monitoring reports). The company is committed to continuous improvement of staff capacity on HSE issues through training and to close collaboration with the O&M contractor, government regulatory bodies, and host communities so as to ensure smooth operation of the wind farm facilities. With a view to improvement, TBCC is advised to continue submitting annual monitoring reports to ADB and to provide updates on the (i) implementation of measures to improve the waste management plan, (ii) installation of safety signage and barricades around the remaining excavated lands, (iii) an ornithologist's recommendation regarding restoration of the excavated land to prevent bird congregation, and (iv) assessment and planting of guggul as per the revised flora and fauna management plan.

SOCIAL IMPACT

A. Introduction

1. The Triconboston Wind Power Project comprises the construction, commissioning, and operation of three adjacent 50-megawatt wind power projects in Thatta District, Sindh Province, south of Pakistan. The borrower is Triconboston Consulting Corporation (TBCC), a special purpose vehicle set up by the project sponsors, Sapphire Group and Bank Alfalah Limited (BAFL). Sapphire Group is a leading local conglomerate with experience in thermal and renewable power project development. BAFL is a publicly listed commercial bank that operates in more than 200 cities in Pakistan and seven cities overseas. The International Finance Corporation holds a 15% stake in BAFL and is also a lender of the project. The project is classified as category C for involuntary resettlement and category C for indigenous peoples, pursuant to the Safeguards Policy Statement (2009) of the Asian Development Bank (ADB). This appendix is an evaluation of the project's implementation and social compliance performance against ADB's safeguards and social requirements.

B. Review Findings

2. **Involuntary Resettlement and Indigenous Peoples.** The wind farm is located on 3,852 acres of land owned by the Government of Sindh. The property was leased to the Alternate Energy Development Board, which in turn leased it to TBCC. All the project's components, including 87 turbines, substations, and internal access roads, are located within the leased land.

3. The initial environmental examination prepared in 2016 confirmed that in the last few decades on the majority of project land there has been no (i) agriculture activity, (ii) habitat of permanent existence, (iii) commercial activity to support the livelihood of residents of nearby villages, (iv) human settlement, or (v) green field or wetland or sanctuary. The social due diligence identified a number of households using some portion of the project site for opportunistic grazing when occasional rain results in the growth of bushes. As the project site remained accessible and other grazing grounds were still available, the project did not affect these occasional herders. The 7-kilometer transmission line was built by the National Transmission and Dispatch Company on government land and did not lead to any involuntary resettlement impact on any individual or household.

4. Sindh is home to the Sindhis, the majority of which are Muslims. Sindhi and Urdu are the most widely spoken languages in the province. In Thatta District, specifically, Sindhi is spoken by most of the population. The project did not affect any indigenous peoples. Annual environmental and monitoring reports by TBCC indicated that there were no additional land acquisitions for the project or any unanticipated impacts on involuntary resettlement and indigenous peoples.

5. **Stakeholder Engagement and Grievance Redress Mechanism**. The project developed a stakeholder engagement plan (SEP) to guide the consultation process. The SEP recognized that engagement should be a broad, inclusive, and continuous process throughout the project life cycle. The SEP provided steps and techniques for stakeholder identification and analysis as well as indicated the resources required. As part of the initial environmental examination, public consultations and various meetings were conducted. Local villagers were positive about the project as they anticipated employment opportunities and more public infrastructure support.

6. The SEP also outlined the need for effective grievance management as essential to maintaining good stakeholder relations. The project established both an external grievance

mechanism and an internal grievance mechanism with each mechanism governed by a separate policy. TBCC's administration manager is responsible for receiving and handling the grievances from both mechanisms. The annual monitoring reports confirmed that the grievance mechanisms and procedures are still in place, reporting a total of 39 grievances, of which 27 are internal and 12 are external from 2017 to 2021. All the grievances are closed.[1]

7. TBCC reported annually on the stakeholder engagement meetings conducted. Nine formal stakeholder meetings were held from 2017 to 2021 in addition to various informal engagements. However, as the pandemic hit, TBCC had to rely more on informal engagements to gather feedback from the community. The SEP will be amended, based on this experience, to define an alternative method of consultation focusing on individual interactions.

8. **Labor.** TBCC ensures its contractors and subcontractors' compliance with national labor laws through formal contracts, binding them to comply with Pakistan Labor laws including minimum wage, child labor, forced labor, harassment, working hours, etc. for their and their sub-contractor's activities. To ensure compliance with requirements of ADB and the International Finance Corporation, the environmental and social action plan (ESAP) identified measures to further strengthen TBCC's labor management policies, which the environmental and social management plan (ESMP) elaborated on. Among others, TBCC developed policies on the following matters: hiring-recruitment and prohibition of child labor, prohibition of forced labor, prohibition of discrimination, freedom of association and collective bargaining, as well as policies on prohibition of harassment and abuse, compensation and benefits, and hours of work. TBCC continuously reported on its and its contractors' compliance with requirements on child and forced labor. TBCC also reported the changes required in its human resource policies following the results of its internal audit. Furthermore, TBCC provided updates on the implementation of the EPC contractor's hiring plan making sure that workers are informed of the time-bound requirement of the project.

9. **Corporate social responsibility activities**. TBCC developed a corporate social responsibility (CSR) policy that emphasized its commitment to being a good neighbor and partner in the community and providing a fair return to society, its employees, and other stakeholders. Among the activities conducted from 2017 to 2021 were (i) construction of water storage tanks in all 17 neighboring villages, (ii) provision of drinking water to all 17 neighboring villages, and (iii) providing access to medical facilities to local villagers.

C. Conclusion and Recommendation

10. Based on the review of safeguards monitoring reports and relevant project documents, there are no outstanding social safeguard compliance issues or claims. The social action items identified in the ESAP were completed, including the development of human resource policies and procedures to protect the workforce, continuation of stakeholder consultation during all phases of the project, and strengthening of the grievance redress mechanism. TBCC completed and submitted to ADB four annual environmental and social monitoring reports covering the period from July 2017 to June 2021. Overall, TBCC's social safeguards performance is deemed satisfactory.

[1] Typical internal grievances related to malfunctioning equipment, and typical external grievances related to water supply to villages.